

Effective Resistance Augmented Monte-Carlo Tree Search: Convergence Analysis for Power Mean and Convex Regularized Planning

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Abstract

We introduce MCTS-ER, a family of Monte-Carlo Tree Search algorithms augmented with an *effective resistance* (ER) bonus derived from the search tree. The ER bonus provides a path-level uncertainty signal: it penalizes trajectories containing weakly-visited edges, complementing local exploration–exploitation mechanisms. We study MCTS-ER in two frameworks. **Part I:** We augment the polynomial exploration bonus of Stochastic-Power-UCT and prove that MCTS-ER retains the same $O(n^{-1/2})$ polynomial convergence rate for root value estimation. **Part II:** We augment convex-regularized MCTS (MENTS, RENTS, TENTS) with the ER bonus and prove that the exponential convergence rate $O(t \exp(-t/(\log t)^3))$ is preserved. For the Tsallis entropy case (TENTS-ER), we show that the ER bonus dynamically modulates the sparsity set of the Tsallis policy, providing adaptive exploration that broadens the active action set in the early phase and contracts to sparse exploitation as visits accumulate. We derive regret bounds and error analysis for each regularizer augmented with ER, showing that ER comes “for free” in terms of asymptotic rates while providing structurally motivated finite-time improvements.

1 Introduction

Monte-Carlo Tree Search (MCTS) is a powerful online planning paradigm that constructs a search tree by combining Monte-Carlo sampling with forward tree expansion [Browne et al., 2012]. The Upper Confidence bound for Trees (UCT) algorithm [Kocsis and Szepesvári, 2006], which adapts UCB1 [Auer et al., 2002] to tree search, is the most widely used selection strategy. However, the theoretical analysis of UCT suffers from issues related to the logarithmic bonus term [Shah et al., 2022], leading to the development of Fixed-Depth-MCTS with a polynomial bonus [Shah et al., 2022] and Stochastic-Power-UCT with power mean estimation [Dam et al., 2024]. More recently, convex regularization has been introduced to MCTS through maximum entropy (MENTS) [Xiao et al., 2019], relative entropy (RENTS), and Tsallis entropy (TENTS) regularizers [Dam et al., 2021], achieving exponential convergence rates.

A limitation of all these MCTS selection strategies is that the exploration mechanism at each node depends only on *local* visit counts. This ignores the global structure of the search tree: a node may have a high local visit count but be reachable only through a weakly-explored edge higher in the tree, making the entire trajectory unreliable.

Effective resistance as a path-level uncertainty measure. In spectral graph theory, the *effective resistance* between two nodes in a graph captures how well-connected they are, accounting for all paths between them [Doyle and Snell, 1984, Chandra et al., 1996]. On a tree (where there is exactly one path between any pair of nodes), the effective resistance equals the sum of edge resistances along

that unique path [Ghosh et al., 2008, Tetali, 1991]: for edge resistances $r(e) = 1/N(e)$ (where $N(e)$ is the visit count of edge e), we obtain

$$R_{\text{eff}}(\text{root}, s) = \sum_{e \in \text{path}(\text{root}, s)} \frac{1}{N(e)}.$$

We propose this quantity as a measure of *trajectory-level uncertainty*: a single weakly-visited edge creates a large contribution, flagging the entire path as under-explored.

Contributions. We propose MCTS-ER, which adds the effective resistance bonus to two distinct MCTS frameworks. Our main contributions are:

1. **Part I (Sections 2–4):** We augment Stochastic-Power-UCT with an additive ER bonus and prove $O(n^{-1/2})$ polynomial convergence, showing the ER bonus is asymptotically dominated by the polynomial bonus.
2. **Part II (Sections 5–7):** We augment convex-regularized MCTS (MENTS, RENTS, TENTS) with an ER bonus injected into the Q-values before computing the regularized policy. We prove that the exponential convergence rate is preserved for all three regularizers.
3. For the TENTS-ER case, we show that the ER bonus dynamically controls the sparsity set $\mathcal{K}$ of the Tsallis policy, providing an *adaptive sparsity schedule* that neither component achieves alone.
4. We derive ER-augmented regret bounds and error analysis for each regularizer (Section 7.5).

PART I: POWER MEAN MCTS-ER

2 Preliminaries

2.1 Markov Decision Process

We consider a discrete-time discounted MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, R, P, \gamma \rangle$ where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the finite action space, $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ is the reward function, $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ is the transition kernel, and $\gamma \in [0, 1)$ is the discount factor. The optimal Q-value satisfies the Bellman equation:

$$Q^*(s, a) = \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma \max_{a'} Q^*(s', a')].$$

2.2 MCTS and Stochastic-Power-UCT

We analyze MCTS with a fixed planning depth H and a playout policy π_0 with value V_0 . For a node s_h at depth h , define inductively $\tilde{V}(s_H) = V_0(s_H)$ and

$$\tilde{Q}(s_h, a) = r(s_h, a) + \gamma \sum_{s_{h+1}} P(s_{h+1}|s_h, a) \tilde{V}(s_{h+1}), \quad (1)$$

$$\tilde{V}(s_h) = \max_a \tilde{Q}(s_h, a). \quad (2)$$

After t trajectories, the value and Q-value estimates in Stochastic-Power-UCT are:

$$\hat{V}_t(s_h) = \left(\sum_{a \in \mathcal{A}_{s_h}} \frac{T_{s_h, a}(t)}{t} (\hat{Q}_{T_{s_h, a}(t)}(s_h, a))^p \right)^{1/p}, \quad (3)$$

$$\hat{Q}_t(s_h, a) = \frac{1}{t} \sum_{i=1}^t [r^i(s_h, a) + \gamma \hat{V}_{T_{s_h, a}^{s_{h+1}}(i)}(s_{h+1})], \quad (4)$$

where $p \in [1, +\infty)$ is the power mean parameter.

Definition 1 (Polynomial concentration). A sequence of estimators $(\widehat{V}_n)_{n \geq 1}$ concentrates at rate (α, β) towards a limit V if there exists a constant $c > 0$ such that for all $n \geq 1$ and all $\varepsilon > n^{-\alpha/\beta}$:

$$\mathbb{P}(|\widehat{V}_n - V| > \varepsilon) \leq cn^{-\alpha}\varepsilon^{-\beta}.$$

We write $\widehat{V}_n \xrightarrow{\alpha, \beta} V$.

2.3 Effective Resistance on Trees

Definition 2 (Tree effective resistance). For a search tree $\mathcal{T}$ with root s_0 and edge visit counts $\{N(e)\}_{e \in \mathcal{T}}$, define the effective resistance from the root to a node s_h at depth h as:

$$R_{\text{eff}}(s_0, s_h) = \sum_{e \in \text{path}(s_0, s_h)} \frac{1}{N(e)},$$

where $\text{path}(s_0, s_h)$ is the unique path from s_0 to s_h in $\mathcal{T}$, and $N(e)$ is the number of times edge e has been traversed.

Remark 3 (Local decomposition on trees). On a tree, when selecting action a at node s_h , the effective resistance to the resulting child s_{h+1} decomposes as:

$$R_{\text{eff}}(s_0, s_{h+1}) = \underbrace{R_{\text{eff}}(s_0, s_h)}_{\text{constant across actions}} + \frac{1}{T_{s_h, a}(t)}.$$

Therefore, the ER contribution to action selection at node s_h reduces to the local quantity $1/T_{s_h, a}(t)$. This decomposition is the key structural property enabling our analysis.

3 MCTS-ER Algorithm

MCTS-ER modifies the action selection of Stochastic-Power-UCT by adding an ER bonus term. At depth h , the selected action is:

$$a_h = \arg \max_{a \in \mathcal{A}_{s_h}} \left\{ \widehat{Q}_{T_{s_h, a}(t)}(s_h, a) + B_h^{\text{poly}}(t, s_h, a) + B_h^{\text{ER}}(t, s_h, a) \right\}, \quad (5)$$

where the polynomial bonus is

$$B_h^{\text{poly}}(t, s_h, a) = C_1 \frac{T_{s_h}(t)^{b_{h+1}/\beta_{h+1}}}{T_{s_h, a}(t)^{\alpha_{h+1}/\beta_{h+1}}}, \quad (6)$$

with algorithmic constants $\{b_i\}_{i=0}^H, \{\alpha_i\}_{i=0}^H, \{\beta_i\}_{i=0}^H$ satisfying the conditions of Stochastic-Power-UCT (Table 1), and the ER bonus is

$$B_h^{\text{ER}}(t, s_h, a) = \frac{C_2}{T_{s_h, a}(t)}, \quad (7)$$

where $C_2 > 0$ is a constant controlling the strength of the ER contribution.

By Remark 3, B_h^{ER} is precisely the incremental effective resistance contributed by the edge (s_h, a) . The combined bonus is therefore:

$$B_h(t, s_h, a) = C_1 \frac{T_{s_h}(t)^{b_{h+1}/\beta_{h+1}}}{T_{s_h, a}(t)^{\alpha_{h+1}/\beta_{h+1}}} + \frac{C_2}{T_{s_h, a}(t)}. \quad (8)$$

The value backup uses the same power mean operator as in Stochastic-Power-UCT (Eq. (3)).

4 Theoretical Analysis

Planning in MCTS requires a sequence of decisions along the tree, with each internal node acting as a non-stationary bandit. The empirical mean at these nodes shifts due to the action selection strategy. Following Dam et al. [2024], we first analyze non-stationary multi-armed bandit settings with the combined ER bonus, focusing on the concentration properties of the power-mean backup for each arm compared to the optimal value. We then apply these findings to MCTS.

Table 1: Conditions for algorithmic constants. $i \in [0, H]$.

Algorithmic constants (all conditions must hold simultaneously)	
$b_i < \alpha_i;$	$b_i > 2.$
$1 \leq p \leq 2; \alpha_i \leq \frac{\beta_i}{2}$	OR $p > 2; \alpha_i \leq \frac{\beta_i}{2}; 0 < \alpha_i - \frac{\beta_i}{p} < 1.$
$\alpha_i(1 - \frac{b_i}{\alpha_i}) \leq b_i < \alpha_i.$	
$\alpha_i = (b_{i+1} - 1)(1 - \frac{b_{i+1}}{\alpha_{i+1}}).$	
$\beta_i = (b_{i+1} - 1).$	

Algorithm 1 MCTS-ER: Effective Resistance Augmented Stochastic-Power-UCT

Require: Root state s_0 , depth H , constants C_1, C_2 , power mean p , playout policy π_0

- 1: Initialize tree $\mathcal{T}$ with root s_0
- 2: **for** $t = 1, 2, \dots, n$ **do**
- 3: **Selection:** Starting from s_0 , at each depth h :

$$a_h = \arg \max_a \left\{ \hat{Q}_{T_{s_h, a}}(s_h, a) + C_1 \frac{T_{s_h, a}^{b_{h+1}/\beta_{h+1}}}{T_{s_h, a}^{\alpha_{h+1}/\beta_{h+1}}} + \frac{C_2}{T_{s_h, a}} \right\}$$

- 4: **Expansion:** If s_{ℓ_t} is a leaf with $\ell_t < H$, add children.
 - 5: **Evaluation:** $\hat{V}(s_{\ell_t}) = \pi_0(s_{\ell_t})$ (playout).
 - 6: **Backup:** Update $\hat{Q}$ via Eq. (4) and $\hat{V}$ via Eq. (3) along trajectory.
 - 7: **end for**
 - 8: **return** $\hat{a} = \arg \max_a \hat{Q}_{T_{s_0, a}(n)}(s_0, a)$
-

4.1 Non-Stationary Power Mean Multi-Armed Bandit with ER Bonus

We consider a class of non-stationary multi-armed bandit (MAB) problems. Let $K \geq 1$ arms and let $X_{a,t}$ denote the random reward obtained by playing arm $a \in [K]$ at time step t , with rewards bounded in $[0, R]$. We write $\hat{\mu}_{a,n} = \frac{1}{n} \sum_{t=1}^n X_{a,t}$ for the average reward collected at arm a after n plays, and $\mu_{a,n} = \mathbb{E}[\hat{\mu}_{a,n}]$. We assume the arms satisfy Assumption 1 (polynomial concentration as in Definition 1).

Let $\mu^* = \max_{a \in [K]} \{\mu_a\}$ be the optimal mean value (assumed unique) and $\Delta_a = \mu^* - \mu_a$ the sub-optimality gap. The MCTS-ER action selection strategy at each time step $n > K$ is:

$$a_n = \arg \max_{a \in \{1, \dots, K\}} \left\{ \hat{\mu}_{a, T_a(n)} + C_1 \frac{n^{b/\beta}}{T_a(n)^{\alpha/\beta}} + \frac{C_2}{T_a(n)} \right\}, \quad (9)$$

where $T_a(n) = \sum_{t=1}^{n-1} \mathbf{1}(a_t = a)$ denotes the number of selections of arm a prior to round n . The combined upper confidence index is

$$U_{a,n,s}^{\text{ER}} := \hat{\mu}_{a,s} + C_1 \frac{n^{b/\beta}}{s^{\alpha/\beta}} + \frac{C_2}{s}. \quad (10)$$

Given a constant $1 \leq p < \infty$, the power mean backup operator is defined as

$$\hat{\mu}_n(p) = \left(\sum_{a=1}^K \frac{T_a(n)}{n} \hat{\mu}_{a, T_a(n)}^p \right)^{1/p}.$$

4.2 Dominance of the Polynomial Bonus

Lemma 4 (ER bonus is asymptotically dominated). *Let $\alpha/\beta \leq 1/2$ and $b/\alpha < 1$ (conditions from Table 1). For the combined bonus with parent visit count n and child visit count s :*

$$B(n, s) = C_1 \frac{n^{b/\beta}}{s^{\alpha/\beta}} + \frac{C_2}{s}.$$

Then for all $s \geq 1$:

$$\frac{C_2}{s} \leq C_1 \frac{n^{b/\beta}}{s^{\alpha/\beta}} \quad \text{whenever } s \leq \left(\frac{C_1}{C_2} \right)^{\beta/(\alpha-\beta)} n^{b/(\alpha-\beta)}.$$

Moreover, since $\alpha/\beta \leq 1/2 < 1$, we have $1/s = o(1/s^{\alpha/\beta})$ as $s \rightarrow \infty$, so the ER bonus decays strictly faster than the polynomial bonus.

Proof. The inequality $C_2/s \leq C_1 n^{b/\beta}/s^{\alpha/\beta}$ is equivalent to $C_2 s^{\alpha/\beta-1} \leq C_1 n^{b/\beta}$. Since $\alpha/\beta \leq 1/2 < 1$, we have $\alpha/\beta - 1 < 0$, so $s^{\alpha/\beta-1}$ is decreasing in s . For any $s \geq 1$, we get $s^{\alpha/\beta-1} \leq 1 \leq C_1 n^{b/\beta}/C_2$ for sufficiently large n .

For the asymptotic claim, note that $s^{-1}/s^{-\alpha/\beta} = s^{\alpha/\beta-1} \rightarrow 0$ as $s \rightarrow \infty$ since $\alpha/\beta < 1$. $\square$

4.3 Modified Concentration Lemmas

We now introduce the ER-modified threshold and establish the concentration properties that parallel Lemmas 4–6 of Dam et al. [2024].

Definition 5 (ER-modified threshold). *For each sub-optimal arm a , define*

$$A_a^{\text{ER}}(n) := \inf \left\{ s \leq n : C_1 \frac{n^{b/\beta}}{s^{\alpha/\beta}} + \frac{C_2}{s} \leq \frac{\Delta_a}{2} \right\}. \quad (11)$$

Proposition 6 (Threshold comparison). *The modified threshold satisfies*

$$A_a^{\text{ER}}(n) \leq A_a(n) + D_a, \quad \text{where } D_a = \frac{4C_2}{\Delta_a}, \quad (12)$$

and $A_a(n) = \left(\frac{2C_1}{\Delta_a} \right)^{\beta/\alpha} n^{b/\alpha}$ is the original threshold of Dam et al. [2024]. In particular, $A_a^{\text{ER}}(n) = A_a(n)(1 + o(1))$ as $n \rightarrow \infty$.

Proof. At $s = A_a(n)$, the polynomial bonus satisfies $C_1 n^{b/\beta} s^{-\alpha/\beta} = \Delta_a/2$. The ER bonus contributes at most $C_2/A_a(n)$. We need $C_1 n^{b/\beta}/s^{\alpha/\beta} + C_2/s \leq \Delta_a/2$. Taking a first-order Taylor expansion of the polynomial bonus around $A_a(n)$, the derivative with respect to s is $-\frac{\alpha}{\beta} C_1 n^{b/\beta} s^{-\alpha/\beta-1}$. The shift required to absorb the ER residual $C_2/A_a(n)$ is:

$$D_a \leq \frac{C_2/A_a(n)}{\frac{\alpha}{\beta} C_1 n^{b/\beta} A_a(n)^{-\alpha/\beta-1}} = \frac{C_2 \beta}{\alpha \cdot \Delta_a/2} = \frac{2C_2 \beta}{\alpha \Delta_a} \leq \frac{4C_2}{\Delta_a},$$

using $\beta/\alpha \leq 2$ (from the condition $\alpha \leq \beta/2$). Since D_a is independent of n and $A_a(n) = \Theta(n^{b/\alpha}) \rightarrow \infty$, we get $A_a^{\text{ER}}(n)/A_a(n) \rightarrow 1$. $\square$

Throughout we write $A^{\text{ER}}(n) := \left(\frac{2C_1}{\Delta} \right)^{\beta/\alpha} n^{b/\alpha} + D$ where $\Delta = \min_{a: \mu_a < \mu^*} \Delta_a$ and $D = \frac{4C_2}{\Delta}$.

Lemma 7 (Modified Lemma 4 — Concentration at optimal arm). *Consider the bandit problem with MCTS-ER selection (9). With $R \geq \varepsilon \geq n^{-\alpha/\beta}$:*

$$\mathbb{P}(|\hat{\mu}_{a^*, T_{a^*}(n)} - \mu^*| > \varepsilon) \leq \sum_{a \neq a^*} \mathbb{P}(T_a(n) > A_a^{\text{ER}}(n) + 1) + \frac{c}{\alpha - 1} \varepsilon^{-\beta} (n - (K-1)A^{\text{ER}}(n) - 1)^{-\alpha+1}. \quad (13)$$

Proof. Define the event $E := \{\sum_{a \neq a^*} T_a(n) > (K-1)(A_a^{\text{ER}}(n) + 1)\}$. Then

$$\mathbb{P}(|\hat{\mu}_{a^*, T_{a^*}(n)} - \mu^*| > \varepsilon) \leq \mathbb{P}(E) + \mathbb{P}(E^c; |\hat{\mu}_{a^*, T_{a^*}(n)} - \mu^*| > \varepsilon).$$

When E^c holds, $T_{a^*}(n) = n - \sum_{a \neq a^*} T_a(n) \geq n - (K-1)(A^{\text{ER}}(n) + 1)$. By a union bound over $a \neq a^*$, $\mathbb{P}(E) \leq \sum_{a \neq a^*} \mathbb{P}(T_a(n) > A_a^{\text{ER}}(n) + 1)$. For the second term, using Assumption 1:

$$\begin{aligned} \mathbb{P}(E^c; |\hat{\mu}_{a^*, T_{a^*}(n)} - \mu^*| > \varepsilon) &\leq \sum_{t=n-(K-1)(A^{\text{ER}}(n)+1)}^n \mathbb{P}(|\hat{\mu}_{a^*, t} - \mu^*| \geq \varepsilon) \\ &\leq \sum_{t=n-(K-1)(A^{\text{ER}}(n)+1)}^n ct^{-\alpha} \varepsilon^{-\beta} \\ &\leq c\varepsilon^{-\beta} \int_{n-(K-1)(A^{\text{ER}}(n)+1)-1}^{\infty} t^{-\alpha} dt \\ &= \frac{c}{\alpha-1} \varepsilon^{-\beta} (n - (K-1)(A^{\text{ER}}(n) + 1) - 1)^{-\alpha+1}, \end{aligned}$$

where we used $\alpha > 2$ (from the conditions in Table 1). $\square$

Lemma 8 (Modified Lemma 5 — Concentration of combined UCB). *Under Assumption 1, for the combined index (10):*

- (i) For all $s \in \{1, \dots, n\}$: $\mathbb{P}(U_{a,n,s}^{\text{ER}} < \mu_a) \leq cC_1^{-\beta} n^{-b}$.
- (ii) For all $s \in \{A_a^{\text{ER}}(n), \dots, n\}$: $\mathbb{P}(U_{a,n,s}^{\text{ER}} > \mu^*) \leq cC_1^{-\beta} n^{-b}$.

Proof. Part (i). Since the ER bonus $C_2/s \geq 0$, we have $U_{a,n,s}^{\text{ER}} \geq U_{a,n,s} = \hat{\mu}_{a,s} + C_1 n^{b/\beta} s^{-\alpha/\beta}$. Therefore:

$$\mathbb{P}(U_{a,n,s}^{\text{ER}} < \mu_a) \leq \mathbb{P}(U_{a,n,s} < \mu_a) = \mathbb{P}\left(\hat{\mu}_{a,s} - \mu_a < -C_1 \frac{n^{b/\beta}}{s^{\alpha/\beta}}\right) \leq cC_1^{-\beta} n^{-b},$$

where the last step uses Assumption 1.

Part (ii). For $s \geq A_a^{\text{ER}}(n)$, by definition $C_1 n^{b/\beta} s^{-\alpha/\beta} + C_2 s^{-1} \leq \Delta_a/2$. We compute:

$$\begin{aligned} \mathbb{P}(U_{a,n,s}^{\text{ER}} > \mu^*) &= \mathbb{P}\left(\hat{\mu}_{a,s} + C_1 \frac{n^{b/\beta}}{s^{\alpha/\beta}} + \frac{C_2}{s} > \mu^*\right) \\ &= \mathbb{P}\left(\hat{\mu}_{a,s} - \mu_a > \Delta_a - C_1 \frac{n^{b/\beta}}{s^{\alpha/\beta}} - \frac{C_2}{s}\right) \\ &\leq \mathbb{P}\left(\hat{\mu}_{a,s} - \mu_a > \frac{\Delta_a}{2}\right). \end{aligned}$$

Because $s \geq A_a^{\text{ER}}(n)$, we also have $C_1 n^{b/\beta} s^{-\alpha/\beta} \leq C_1 n^{b/\beta} s^{-\alpha/\beta} + C_2/s \leq \Delta_a/2$, so:

$$\mathbb{P}\left(\hat{\mu}_{a,s} - \mu_a > \frac{\Delta_a}{2}\right) \leq \mathbb{P}\left(\hat{\mu}_{a,s} - \mu_a > C_1 \frac{n^{b/\beta}}{s^{\alpha/\beta}}\right) \leq cC_1^{-\beta} n^{-b}. \quad \square$$

Lemma 9 (Modified Lemma 6 — Sub-optimal arm count). *Assume $b > 1$. For all $u \geq A_a^{\text{ER}}(n)$:*

$$\mathbb{P}(T_a(n) \geq u) \leq \frac{2cC_1^{-\beta}}{b-1} (u-1)^{-(b-1)}.$$

Proof. For any $\tau \in \mathbb{R}$, define the events:

$$E_1 = \{\forall t \in [u, n] : U_{a,t,u}^{\text{ER}} \leq \tau\}, \quad E_2 = \{\forall t' \in [1, n-u] : U_{a^*, u+t', t'}^{\text{ER}} > \tau\}.$$

We claim $E_1 \cap E_2 \Rightarrow T_a(n) \leq u$. Suppose for contradiction that $T_a(n) > u$. Let t' denote the time at which arm a is played for the $(u+1)$ -th time. At any time t with $u+t' \leq t \leq n$, arm a has been played exactly u times and the optimal arm has been played t' times, with $t' \leq n-u$. Under $E_1 \cap E_2$:

$$U_{a^*, t, t'}^{\text{ER}} \geq U_{a^*, u+t', t'}^{\text{ER}} > \tau \geq U_{a, t, u}^{\text{ER}},$$

where the first inequality uses that $U_{a^*,t,t'}^{\text{ER}}$ is non-decreasing in the first argument (since $n^{b/\beta}$ is increasing in n). This means arm a would not be selected after u plays, contradicting $T_a(n) > u$.

Therefore $\{T_a(n) \geq u\} \subseteq E_1^c \cup E_2^c$, giving:

$$\mathbb{P}(T_a(n) \geq u) \leq \sum_{t=u}^n \mathbb{P}(U_{a,t,u}^{\text{ER}} > \mu^*) + \sum_{t'=1}^{n-u} \mathbb{P}(U_{a^*,u+t',t'}^{\text{ER}} \leq \mu^*),$$

where we set $\tau = \mu^*$. Since $u \geq A_a^{\text{ER}}(n)$, Lemma 8(ii) yields:

$$\sum_{t=u}^n \mathbb{P}(U_{a,t,u}^{\text{ER}} > \mu^*) \leq cC_1^{-\beta} \sum_{t=u}^n t^{-b} \leq cC_1^{-\beta} \int_{u-1}^{\infty} t^{-b} dt = \frac{cC_1^{-\beta}}{b-1} (u-1)^{-(b-1)}.$$

Similarly, by Lemma 8(i):

$$\sum_{t'=1}^{n-u} \mathbb{P}(U_{a^*,u+t',t'}^{\text{ER}} \leq \mu^*) \leq cC_1^{-\beta} \sum_{t'=1}^{n-u} (u+t')^{-b} \leq \frac{cC_1^{-\beta}}{b-1} (u-1)^{-(b-1)}.$$

Combining the two bounds yields $\mathbb{P}(T_a(n) \geq u) \leq \frac{2cC_1^{-\beta}}{b-1} (u-1)^{-(b-1)}$. $\square$

4.4 Power Mean Error Decomposition with ER Bonus

Lemma 10 (Power mean error bound — Lemma 7 of Dam et al. [2024]). *For any $p \geq 1$:*

$$|\hat{\mu}_n(p) - \mu^*| \leq R \sum_{a \neq a^*} \frac{T_a(n)}{n} + \left(\sum_{a=1}^K \frac{T_a(n)}{n} |\hat{\mu}_{a,T_a(n)} - \mu_a|^p \right)^{1/p}. \quad (14)$$

Proof. We provide the full proof as it is the structural starting point for the ER analysis.

Upper bound. Since $\hat{\mu}_{a,T_a(n)} \leq \mu_a + |\hat{\mu}_{a,T_a(n)} - \mu_a|$ and $\mu^* = \max_a \mu_a$, applying Minkowski's inequality (Lemma 3 of Dam et al. [2024]):

$$\begin{aligned} \hat{\mu}_n(p) - \mu^* &\leq \frac{(\sum_a T_a(n) |\hat{\mu}_{a,T_a(n)}|^p)^{1/p} - (\sum_a T_a(n) (\mu_a)^p)^{1/p}}{n^{1/p}} \\ &\leq \frac{(\sum_a T_a(n) |\mu_a + |\hat{\mu}_{a,T_a(n)} - \mu_a||^p)^{1/p} - (\sum_a T_a(n) (\mu_a)^p)^{1/p}}{n^{1/p}} \\ &\leq \left(\sum_{a=1}^K \frac{T_a(n)}{n} |\hat{\mu}_{a,T_a(n)} - \mu_a|^p \right)^{1/p}. \end{aligned}$$

Lower bound. Since $\mu_a \leq \hat{\mu}_{a,T_a(n)} + |\hat{\mu}_{a,T_a(n)} - \mu_a|$ and the power mean is increasing in p :

$$\begin{aligned} \mu^* - \hat{\mu}_n(p) &= \frac{n\mu^* - n\hat{\mu}_n(p)}{n} = \frac{\sum_{a \neq a^*} T_a(n) (\mu^* - \mu_a) + \sum_a T_a(n) \mu_a - n\hat{\mu}_n(p)}{n} \\ &\leq R \sum_{a \neq a^*} \frac{T_a(n)}{n} + \left(\sum_{a=1}^K \frac{T_a(n)}{n} (\mu_a)^p \right)^{1/p} - \hat{\mu}_n(p). \end{aligned}$$

The second difference is bounded by Minkowski's inequality applied in the reverse direction, yielding the same $(\sum_a \frac{T_a}{n} |\hat{\mu}_a - \mu_a|^p)^{1/p}$ term. Combining gives (14). $\square$

4.5 Modified Concentration of Optimal and Sub-optimal Arms

Lemma 11 (Modified Lemma 8 — Optimal arm concentration). *With $R \geq \varepsilon \geq n^{-\alpha/\beta}$:*

$$\mathbb{P}\left(\frac{T_{a^*}(n)}{n} |\hat{\mu}_{a^*,T_{a^*}(n)} - \mu^*|^p > \varepsilon^p\right) \leq \frac{2cC_1^{-\beta}(K-1)}{b-1} A^{\text{ER}}(n)^{-(b-1)} + \frac{c}{\alpha-1} \varepsilon^{-\beta} (n - (K-1)(A^{\text{ER}}(n)+1) - 1)^{-\alpha+1}. \quad (15)$$

Proof. Since $T_{a^*}(n)/n \leq 1$, we have

$$\mathbb{P}\left(\frac{T_{a^*}(n)}{n}|\hat{\mu}_{a^*, T_{a^*}(n)} - \mu^*|^p > \varepsilon^p\right) \leq \mathbb{P}(|\hat{\mu}_{a^*, T_{a^*}(n)} - \mu^*| > \varepsilon).$$

Applying Lemma 7:

$$\begin{aligned} & \mathbb{P}(|\hat{\mu}_{a^*, T_{a^*}(n)} - \mu^*| > \varepsilon) \\ & \leq \sum_{a \neq a^*} \mathbb{P}(T_a(n) > A_a^{\text{ER}}(n) + 1) + \frac{c}{\alpha - 1} \varepsilon^{-\beta} (n - (K - 1)(A^{\text{ER}}(n) + 1) - 1)^{-\alpha+1}. \end{aligned}$$

By Lemma 9 with $u = A_a^{\text{ER}}(n) + 1$:

$$\sum_{a \neq a^*} \mathbb{P}(T_a(n) > A_a^{\text{ER}}(n) + 1) \leq \sum_{a \neq a^*} \frac{2cC_1^{-\beta}}{b-1} A_a^{\text{ER}}(n)^{-(b-1)} \leq \frac{2cC_1^{-\beta}(K-1)}{b-1} A^{\text{ER}}(n)^{-(b-1)}.$$

□

Lemma 12 (Modified Lemma 9 — Sub-optimal arm concentration). *For a sub-optimal arm a , with $R \geq \varepsilon \geq n^{-\alpha/\beta}$, there exists N_0 such that $\forall n \geq N_0$:*

(i) *If $1 \leq p \leq 2$ and $\alpha \leq \beta/p$:*

$$\mathbb{P}\left(\frac{T_a(n)}{n}|\hat{\mu}_{a, T_a(n)} - \mu_a|^p > \frac{\varepsilon^p}{K-1}\right) \leq \frac{2cC_1^{-\beta}}{(b-1)} A^{\text{ER}}(n)^{-(b-1)} + \frac{2c(K-1)^{\beta/p}}{-(\alpha - \beta/p) + 1} \varepsilon^{-\beta} (A_a^{\text{ER}}(n) + 1)^{-(\alpha-1)}.$$

(ii) *If $p > 2$ and $0 < \alpha - \beta/p < 1$:*

$$\mathbb{P}\left(\frac{T_a(n)}{n}|\hat{\mu}_{a, T_a(n)} - \mu_a|^p > \frac{\varepsilon^p}{K-1}\right) \leq \frac{2cC_1^{-\beta}}{(b-1)} A^{\text{ER}}(n)^{-(b-1)} + \frac{c(K-1)^{\beta/p}}{1 - (\alpha - \beta/p)} \varepsilon^{-\beta} (A^{\text{ER}}(n) + 1)^{-(\alpha-1)}.$$

Proof. We split on the event $\{T_a(n) > A_a^{\text{ER}}(n) + 1\}$:

$$\begin{aligned} & \mathbb{P}\left(\frac{T_a(n)}{n}|\hat{\mu}_{a, T_a(n)} - \mu_a|^p > \frac{\varepsilon^p}{K-1}\right) \\ & \leq \mathbb{P}(T_a(n) > A_a^{\text{ER}}(n) + 1) + \sum_{t=1}^{A_a^{\text{ER}}(n)+1} \mathbb{P}\left(\frac{t}{n}|\hat{\mu}_{a, t} - \mu_a|^p > \frac{\varepsilon^p}{K-1}\right). \end{aligned}$$

The first term is bounded by $\frac{2cC_1^{-\beta}}{(b-1)} A^{\text{ER}}(n)^{-(b-1)}$ via Lemma 9. For the second term (call it G_2), we can find N_0 such that for $t \leq A^{\text{ER}}(n) + 1$ and $n \geq N_0$, the condition $(\frac{n}{t(K-1)})^{1/p} \varepsilon > n^{-\alpha/\beta}$ holds. Therefore:

$$G_2 \leq \sum_{t=1}^{A_a^{\text{ER}}(n)+1} ct^{-\alpha} \left(\frac{n}{t(K-1)}\right)^{-\beta/p} \varepsilon^{-\beta} = c(K-1)^{\beta/p} \varepsilon^{-\beta} n^{-\beta/p} \sum_{t=1}^{A_a^{\text{ER}}(n)+1} t^{-(\alpha-\beta/p)}.$$

Case (i): When $\alpha - \beta/p \leq 0$ (which holds for $1 \leq p \leq 2$ since $\alpha \leq \beta/2$), the sum is bounded by integrating $t^{-(\alpha-\beta/p)}$ which is increasing. The dominant term is $(A_a^{\text{ER}}(n) + 1)^{-(\alpha-\beta/p)+1} / (-(\alpha - \beta/p) + 1)$. Since $-(\alpha - \beta/p) + 1 \geq 1$, for $n \geq N_0$:

$$G_2 \leq \frac{2c(K-1)^{\beta/p}}{-(\alpha - \beta/p) + 1} \varepsilon^{-\beta} (A_a^{\text{ER}}(n) + 1)^{-(\alpha-1)}.$$

Case (ii): When $0 < \alpha - \beta/p < 1$, we similarly bound the sum using an integral, obtaining:

$$G_2 \leq \frac{c(K-1)^{\beta/p}}{1 - (\alpha - \beta/p)} \varepsilon^{-\beta} (A^{\text{ER}}(n) + 1)^{-(\alpha-1)}.$$

Note that $A^{\text{ER}}(n)$ appears with the same exponent $-(\alpha - 1)$ as in the original analysis; the additive shift D does not alter the polynomial rate. □

4.6 Main Bandit Result: Power Mean Concentration with ER

Lemma 13 (Modified Lemma 10 — Power mean concentration with ER). *Define $\varepsilon_0 = \frac{2^{1/p}n\varepsilon}{x} + \frac{nR(K-1)}{x} \left(\frac{2^{1/p}(3+A^{\text{ER}}(n))x}{n} \right)$ with $R \geq \varepsilon \geq n^{-\alpha/\beta}$. There exists $N_{\text{ER}} \geq 1$ such that for all $n \geq N_{\text{ER}}$ and $x \geq 1$:*

$$\mathbb{P}\left(|\hat{\mu}_n(p) - \mu^*| \geq \frac{\varepsilon_0 x}{n}\right) \leq \frac{8cC_1^{-\beta}KR^\beta\varepsilon^{-\beta}A^{\text{ER}}(n)^{-(b-1)}}{b-1} + \frac{2cC_1^{-\beta}(K-1)(2^{1/p}(3+A^{\text{ER}}(n))x-1)^{-(b-1)}}{b-1}. \quad (16)$$

Proof. By Lemma 10, $|\hat{\mu}_n(p) - \mu^*| \leq R \sum_{a \neq a^*} \frac{T_a(n)}{n} + \left(\sum_a \frac{T_a(n)}{n} |\hat{\mu}_{a,T_a(n)} - \mu_a|^p \right)^{1/p}$.

Since $\frac{\varepsilon_0 x}{n} = 2^{1/p}\varepsilon + R(K-1) \left(\frac{2^{1/p}(3+A^{\text{ER}}(n))x}{n} \right)$, we decompose:

$$\begin{aligned} \mathbb{P}\left(|\hat{\mu}_n(p) - \mu^*| > \frac{\varepsilon_0 x}{n}\right) &\leq \underbrace{\mathbb{P}\left(R \sum_{a \neq a^*} \frac{T_a(n)}{n} > R(K-1) \cdot \frac{2^{1/p}(3+A^{\text{ER}}(n))x}{n}\right)}_{H_1} \\ &\quad + \underbrace{\mathbb{P}\left(\left(\sum_a \frac{T_a(n)}{n} |\hat{\mu}_{a,T_a(n)} - \mu_a|^p\right)^{1/p} \geq 2^{1/p}\varepsilon\right)}_{H_2}. \end{aligned}$$

Bounding H_1 . For $x \geq 1$:

$$\begin{aligned} H_1 &\leq \sum_{a \neq a^*} \mathbb{P}\left(T_a(n) > 2^{1/p}(3+A^{\text{ER}}(n))x\right) \\ &\leq \frac{2cC_1^{-\beta}(K-1)}{b-1} (2^{1/p}(3+A^{\text{ER}}(n))x-1)^{-(b-1)}, \end{aligned}$$

where we applied Lemma 9 noting that $2^{1/p}(3+A^{\text{ER}}(n))x \geq A_a^{\text{ER}}(n)$ for large n .

Bounding H_2 . We split into optimal and sub-optimal arms:

$$H_2 \leq \mathbb{P}\left(\frac{T_{a^*}(n)}{n} |\hat{\mu}_{a^*,T_{a^*}(n)} - \mu^*|^p > \varepsilon^p\right) + \sum_{a \neq a^*} \mathbb{P}\left(\frac{T_a(n)}{n} |\hat{\mu}_{a,T_a(n)} - \mu_a|^p > \frac{\varepsilon^p}{K-1}\right).$$

By Lemma 11 and Lemma 12, since $b-1 < \alpha-1$ and $n^{-\alpha/\beta} \leq \varepsilon \leq R$, there exists N_{ER} such that for $n \geq N_{\text{ER}}$:

$$H_2 \leq \frac{8cC_1^{-\beta}K(R/\varepsilon)^\beta A^{\text{ER}}(n)^{-(b-1)}}{b-1} = \frac{8cC_1^{-\beta}KR^\beta\varepsilon^{-\beta}A^{\text{ER}}(n)^{-(b-1)}}{b-1}.$$

Combining H_1 and H_2 yields (16). $\square$

Theorem 14 (Concentration of power mean with ER — Modified Theorem 1). *For $a \in [K]$, let $(\hat{\mu}_{a,n})_{n \geq 1}$ satisfy Assumption 1. Assume arms are sampled according to (9) with parameters $\alpha, \beta, b, C_1, C_2$, and p, α, β, b satisfy one of:*

- (i) $1 \leq p \leq 2$ and $\alpha \leq \beta/2$;
- (ii) $p > 2$ and $0 < \alpha - \beta/p < 1$.

If $\alpha(1-b/\alpha) \leq b < \alpha$, then $\hat{\mu}_n(p) \xrightarrow{\alpha', \beta'} \mu^$ with $\alpha' = (b-1)(1-b/\alpha)$ and $\beta' = (b-1)$.*

Proof. We use Lemma 13. Set

$$\varepsilon = 2^{1/p}R(2K-1) \left(\frac{A^{\text{ER}}(n)x}{n} \right).$$

Recall $A^{\text{ER}}(n) = \left(\frac{2C_1}{\Delta}\right)^{\beta/\alpha} n^{b/\alpha} + D$ with $D = \frac{4C_2}{\Delta}$.

We verify $\varepsilon \geq \frac{\varepsilon_0 x}{n}$ for n large enough. Computing:

$$\begin{aligned} \varepsilon - \frac{\varepsilon_0 x}{n} &= 2^{1/p} R(2K-1) \frac{A^{\text{ER}}(n)x}{n} - 2^{1/p} R(K-1) \frac{(A^{\text{ER}}(n)+3)x}{n} - 2^{1/p} \varepsilon' \\ &= 2^{1/p} R K \frac{A^{\text{ER}}(n)x}{n} - 2^{1/p} R(K-1) \frac{3x}{n} - 2^{1/p} \varepsilon' \\ &= \underbrace{2^{1/p} R(K-1) \frac{x}{n} (A^{\text{ER}}(n)-3)}_{T_1} + \underbrace{2^{1/p} R \frac{A^{\text{ER}}(n)x}{n}}_{T_2} - 2^{1/p} \varepsilon', \end{aligned}$$

where $\varepsilon' = n^{-\alpha/\beta} x$ satisfies $\varepsilon' \geq n^{-\alpha/\beta}$. Since $A^{\text{ER}}(n) = \Theta(n^{b/\alpha}) \rightarrow \infty$, $T_1 > 0$ for n large. Since $A^{\text{ER}}(n)/n = \Theta(n^{-(1-b/\alpha)})$ and $1 - b/\alpha \leq 1/2 \leq \alpha/\beta$, we have $RA^{\text{ER}}(n)x/n \geq n^{-(1-b/\alpha)} x \geq n^{-\alpha/\beta} x = \varepsilon'$. So $T_2 \geq 0$.

Therefore $\varepsilon \geq \varepsilon_0 x/n$ for all $n \geq N_0$ for some N_0 . By Lemma 13:

$$\begin{aligned} D &:= \mathbb{P}(|\hat{\mu}_n(p) - \mu^*| \geq \varepsilon) \leq \mathbb{P}\left(|\hat{\mu}_n(p) - \mu^*| \geq \frac{\varepsilon_0 x}{n}\right) \\ &\leq \frac{8cC_1^{-\beta} K R^\beta \varepsilon^{-\beta} A^{\text{ER}}(n)^{-(b-1)}}{b-1} + \frac{2cC_1^{-\beta} (K-1) (A^{\text{ER}}(n)x)^{-(b-1)}}{b-1}, \end{aligned}$$

where we used $2^{1/p}(3 + A^{\text{ER}}(n))x - 1 > A^{\text{ER}}(n)x$ for $x \geq 1$.

Now substituting $\varepsilon = 2^{1/p} R(2K-1) A^{\text{ER}}(n)x/n$ and $A^{\text{ER}}(n) = \left(\frac{2C_1}{\Delta}\right)^{\beta/\alpha} n^{b/\alpha} + D$:

$$\begin{aligned} D &\leq \frac{8cC_1^{-\beta} K R^\beta}{b-1} \cdot n^{\beta(1-b/\alpha)} \cdot A^{\text{ER}}(n)^{-(b-1)} \cdot x^{-\beta} \\ &\quad + \frac{2cC_1^{-\beta} (K-1)}{b-1} \cdot A^{\text{ER}}(n)^{-(b-1)} \cdot x^{-(b-1)}. \end{aligned}$$

Since $A^{\text{ER}}(n)^{-(b-1)} = \Theta(n^{-b(b-1)/\alpha})$ (the additive D does not affect the leading order), we obtain:

$$D \leq c_{\text{ER}} n^{-\alpha'} \varepsilon^{-\beta'},$$

with $\alpha' = \min\{\frac{b}{\alpha}(b-1), b-1\} = \frac{b}{\alpha}(b-1)$ and $\beta' = \max\{b-1, \beta\} = b-1$. Using the condition $\alpha(1-b/\alpha) \leq b$, we can set $\alpha' = (b-1)(1-b/\alpha)$ and $\beta' = (b-1)$.

The constant is

$$c_{\text{ER}} = \frac{16cC_1^{-\beta} K}{b-1} \left(\frac{2C_1}{\Delta}\right)^{-\frac{\beta}{\alpha}(b-1)} \left(\frac{1}{2^{1/p} R(2K-1)} \left(\frac{2C_1}{\Delta}\right)^{-\beta/\alpha}\right)^{-\beta} (1 + D/A(n))^{-(b-1)},$$

which for n large converges to the same constant as in Dam et al. [2024, Eq. 58] with a multiplicative correction factor $(1 + D/A(n))^{-(b-1)} \rightarrow 1$. To ensure the bound holds for all $n \geq 1$ (not just $n \geq N_0$), we note that for $1 \leq n < N_0$, $|\hat{\mu}_n(p) - \mu^*| \leq R$. We can choose c_{ER} large enough so that $c_{\text{ER}} n^{-\alpha'} \varepsilon^{-\beta'} \geq 1$ for all $\varepsilon \leq R$ and $n < N_0$, making the bound trivially correct. $\square$

4.7 MCTS Convergence

We now apply the bandit results to the MCTS tree, proceeding by induction as in Dam et al. [2024].

Theorem 15 (Convergence of MCTS-ER). *When we apply the MCTS-ER algorithm (Algorithm 1) with algorithmic constants $\{b_i\}_{i=0}^H$, $\{\alpha_i\}_{i=0}^H$, $\{\beta_i\}_{i=0}^H$ satisfying the conditions in Table 1, we have:*

(i) *For any node s_h at depth h in the tree ($h = 0, 1, \dots, H$):*

$$\hat{V}_n(s_h) \xrightarrow{\alpha_h, \beta_h} \tilde{V}(s_h).$$

(ii) For any node s_h at depth h in the tree ($h = 0, 1, \dots, H - 1$):

$$\widehat{Q}_n(s_h, a) \xrightarrow{\alpha_{h+1}, \beta_{h+1}} \widetilde{Q}(s_h, a), \quad \text{for all } a \in \mathcal{A}_{s_h}.$$

Proof. We prove by induction on the depth H .

Base case ($H = 1$). The state at the root node is s_0 . Let $r^t(s_0, a)$ be the intermediate reward at time t after visiting (s_0, a) , and $s_1 \sim P(\cdot|s_0, a)$. Recall $\widetilde{Q}(s_0, a) = r(s_0, a) + \gamma \sum_{s_1} P(s_1|s_0, a) \widetilde{V}(s_1)$.

Claim (i) at leaves. For any $s_1 \sim P(\cdot|s_0, a)$, $\widehat{V}_n(s_1)$ is the average of i.i.d. calls to the playout policy $\pi_0(s_1)$, so by standard concentration:

$$\widehat{V}_n(s_1) \xrightarrow{\alpha_1, \beta_1} \widetilde{V}(s_1). \quad (17)$$

Claim (ii) at root. From Eq. (4):

$$\widehat{Q}_n(s_0, a) = \frac{1}{n} \sum_{i=1}^n \left[r^i(s_0, a) + \gamma \widehat{V}_{T_{s_0, a}^{s_1}(i)}(s_1) \right].$$

Applying Lemma 1 of Dam et al. [2024] with $X_i = r^i(s_0, a)$ (i.i.d. rewards with mean $r(s_0, a)$), $p = (p_1, \dots, p_M)$ as the transition probabilities of taking action a at state s_0 , and using (17) for each $(Vb_{m,n})_{n \geq 1}$:

$$\widehat{Q}_n(s_0, a) \xrightarrow{\alpha_1, \beta_1} \widetilde{Q}(s_0, a), \quad \text{for all } a \in \mathcal{A}_{s_0}. \quad (18)$$

Critically, Lemma 1 depends only on the concentration properties of child value estimates and the i.i.d. rewards; it does *not* depend on the selection strategy.

Claim (i) at root. At s_0 , the MCTS-ER selection uses the combined bonus (8). The Q-value estimates (18) satisfy Assumption 1 with rate (α_1, β_1) . By Theorem 14, the power mean estimator under MCTS-ER concentrates:

$$\widehat{V}_n(s_0) = \left(\sum_a \frac{T_{s_0, a}(n)}{n} (\widehat{Q}_{T_{s_0, a}(n)}(s_0, a))^p \right)^{1/p} \xrightarrow{\alpha_0, \beta_0} \widetilde{V}(s_0),$$

with $\alpha_0 = (b_0 - 1)(1 - b_0/\alpha_0)$ and $\beta_0 = (b_0 - 1)$ satisfying Table 1. This concludes the base case.

Inductive step ($H - 1 \rightarrow H$). Assume the theorem holds for trees of depth $H - 1$. Consider a tree of depth H .

When we take action a at root s_0 and get $s_1 \sim P(\cdot|s_0, a)$, we enter a subtree of depth $H - 1$. The MCTS-ER selection within this subtree uses the combined bonus with parameters $(\alpha_{h+1}, \beta_{h+1}, b_{h+1})$ at each depth $h \geq 1$, together with the ER bonus $C_2/T_{s_h, a}$ at every depth. By Lemma 4, the ER bonus is dominated at every depth, and by Remark 3, the ER contribution decomposes locally.

By the induction hypothesis, in the subtree rooted at s_1 :

$$\widehat{V}_n(s_h) \xrightarrow{\alpha_h, \beta_h} \widetilde{V}(s_h), \quad \text{for } h = 1, \dots, H, \quad (19)$$

$$\widehat{Q}_n(s_h, a) \xrightarrow{\alpha_{h+1}, \beta_{h+1}} \widetilde{Q}(s_h, a), \quad \text{for } h = 1, \dots, H - 1. \quad (20)$$

Now consider the root node s_0 . Applying Lemma 1 of Dam et al. [2024] with $X_i = r^i(s_0, a)$ and each $(\widehat{V}_{m,n})_{n \geq 1}$ satisfying (19) at rate (α_1, β_1) , where $s_1 \in \{s_m\}$, $s_m \sim P(\cdot|s_0, a)$:

$$\widehat{Q}_n(s_0, a) \xrightarrow{\alpha_1, \beta_1} \widetilde{Q}(s_0, a), \quad \text{for all } a \in \mathcal{A}_{s_0}. \quad (21)$$

At the root s_0 , applying Theorem 14 with the concentration (21) and the power mean value backup operator:

$$\widehat{V}_n(s_0) \xrightarrow{\alpha_0, \beta_0} \widetilde{V}(s_0), \quad (22)$$

with α_0, β_0 satisfying the recursive relations in Table 1.

Combining (19) and (22) concludes (i). Combining (20) and (21) concludes (ii). The results hold for any node in the tree of depth H , completing the induction. $\square$

Theorem 16 (Convergence of expected payoff). *At the root node s_0 , with the best possible parameter tuning:*

$$|\mathbb{E}[\widehat{V}_n(s_0)] - \widetilde{V}(s_0)| \leq O(n^{-1/2}).$$

Proof. Using the convexity of $f(x) = |x|$ and applying Jensen's inequality:

$$\begin{aligned} |\mathbb{E}[\widehat{V}_n(s_0)] - \widetilde{V}(s_0)| &\leq \mathbb{E}[|\widehat{V}_n(s_0) - \widetilde{V}(s_0)|] = \int_0^\infty \mathbb{P}(|\widehat{V}_n(s_0) - \widetilde{V}(s_0)| \geq s) ds \\ &\leq \int_0^{n^{-\alpha_0/\beta_0}} 1 ds + \int_{n^{-\alpha_0/\beta_0}}^\infty c_{\text{ER}} n^{-\alpha_0} s^{-\beta_0} ds \\ &= n^{-\alpha_0/\beta_0} + c_{\text{ER}} n^{-\alpha_0} \left[\frac{s^{-\beta_0+1}}{-\beta_0+1} \right]_{n^{-\alpha_0/\beta_0}}^\infty = \left(\frac{c_{\text{ER}}}{\beta_0-1} + 1 \right) n^{-\alpha_0/\beta_0}. \end{aligned}$$

Because $\alpha_0/\beta_0 \leq 1/2$ (Theorem 14), the best rate is $O(n^{-1/2})$. $\square$

Remark 17 (Same rate, different constant). *MCTS-ER achieves the same $O(n^{-1/2})$ convergence rate as Stochastic-Power-UCT, but with a potentially larger constant $c_{\text{ER}} \geq c_0$. The constant c_{ER} depends on C_2 , K , $\Delta_{\min}$, and the original parameters. The ER bonus provides a finite-time improvement that is not captured by the asymptotic rate.*

Remark 18 (Optimal exploration bonus). *By selecting algorithmic constants from Table 1 such that $\alpha_i/\beta_i = 1/2$ and $b_i/\beta_i = 1/4$ for $i \in [0, H]$, we achieve the optimal rate. This yields the combined exploration bonus:*

$$B_h(n, s, a) = C_1 \frac{n^{1/4}}{T_{s,a}(n)^{1/2}} + \frac{C_2}{T_{s,a}(n)}.$$

For large $T_{s,a}(n)$, the ER term is negligible; for small $T_{s,a}(n)$ (sparse planning regime), the ER term provides a significant additional push for exploration. These findings align with those of Shah et al. [2022] and Dam et al. [2024], but more broadly apply to the ER-augmented power mean estimator.

PART II: CONVEX REGULARIZED MCTS-ER

5 Convex Regularized MCTS Background

We now study a complementary framework: combining the ER bonus with convex-regularized MCTS [Dam et al., 2021], which achieves exponential (rather than polynomial) convergence rates through the Legendre-Fenchel transform.

5.1 Regularized Bellman Operator

Let $\Omega : \Pi \rightarrow \mathbb{R}$ be a strongly convex function on the policy simplex. The *Legendre-Fenchel transform* (convex conjugate) of Ω is $\Omega^* : \mathbb{R}^{|\mathcal{A}|} \rightarrow \mathbb{R}$:

$$\Omega^*(Q_s) := \max_{\pi_s \in \Pi_s} \langle \pi_s, Q_s \rangle - \tau \Omega(\pi_s), \quad (23)$$

where $\tau > 0$ is the regularization temperature. The maximizing argument defines the regularized policy:

$$\nabla \Omega^*(Q_s) = \arg \max_{\pi_s \in \Pi_s} \langle \pi_s, Q_s \rangle - \tau \Omega(\pi_s). \quad (24)$$

Proposition 19 (Properties of Ω^* , cf. Dam et al. (2021)). *Let Ω be strongly convex.*

1. **Lipschitz gradient:** $\nabla \Omega^*$ is Lipschitz continuous with constant $L_\Omega > 0$.
2. **Boundedness:** If $L_\Omega^{\min} \leq \Omega(\pi_s) \leq U_\Omega$ for all $\pi_s \in \Pi_s$, then $\max_a Q_s(a) - \tau U_\Omega \leq \Omega^*(Q_s) \leq \max_a Q_s(a) - \tau L_\Omega^{\min}$.
3. **Contraction:** $\|\Omega^*(Q_1) - \Omega^*(Q_2)\|_\infty \leq \gamma \|Q_1 - Q_2\|_\infty$.

5.2 E3W Sampling Strategy and Entropy Regularizers

The E3W (Extended Empirical Exponential Weight) strategy [Dam et al., 2021] selects actions via:

$$\pi_t(a|s) = (1 - \lambda_s) \nabla \Omega^*(Q_\Omega(s, \cdot)/\tau)(a) + \frac{\lambda_s}{|\mathcal{A}|}, \quad (25)$$

where $\lambda_s = |\mathcal{A}| \epsilon / \log(\sum_a N(s, a) + 1)$ with $\epsilon > 0$. Three choices of Ω yield the MENTS, RENTS, and TENTS algorithms:

- **Maximum entropy (MENTS):** $\Omega(\pi) = \sum_a \pi(a|s) \log \pi(a|s)$,

$$\Omega^*(Q_s) = \tau \log \sum_a e^{Q(s,a)/\tau}, \quad \nabla \Omega^*(Q_s)(a) = \frac{e^{Q(s,a)/\tau}}{\sum_b e^{Q(s,b)/\tau}}.$$

- **Relative entropy (RENTS):** $\Omega(\pi_t) = D_{\text{KL}}(\pi_t \| \pi_{t-1})$,

$$\nabla \Omega^*(Q_s)(a) = \frac{\pi_{t-1}(a|s) e^{Q_t(s,a)/\tau}}{\sum_b \pi_{t-1}(b|s) e^{Q_t(s,b)/\tau}}.$$

- **Tsallis entropy (TENTS):** $\Omega(\pi) = \frac{1}{2}(\|\pi\|_2^2 - 1)$,

$$\nabla \Omega^*(Q_s)(a) = \max \left\{ \frac{Q(s, a)}{\tau} - \frac{\sum_{a' \in \mathcal{K}} Q(s, a')/\tau - 1}{|\mathcal{K}|}, 0 \right\},$$

where $\mathcal{K} \subseteq \mathcal{A}$ is the *sparsity set* of actions receiving nonzero probability.

5.3 Known Convergence Results

Theorem 20 (Exponential convergence, Dam et al. (2021)). *At the root node s with $N(s)$ visitations, for $\epsilon > 0$ and constants $C, \hat{C}$:*

$$\mathbb{P}(|V_\Omega(s) - V_\Omega^*(s)| > \epsilon) \leq C \exp \left\{ -\frac{N(s)}{\hat{C} \sigma \log^2(2 + N(s))} \right\}.$$

Theorem 21 (Exponential action convergence, Dam et al. (2021)). *Let a_t be the action returned by E3W at step t . For large enough t :*

$$\mathbb{P}(a_t \neq a^*) \leq C t \exp \left\{ -\frac{t}{\hat{C} \sigma (\log t)^3} \right\}.$$

6 ER-Augmented Convex Regularized MCTS

We augment the convex-regularized framework by injecting the ER bonus directly into the Q-values before computing the regularized policy.

Definition 22 (ER-augmented Q-values). *At node s with action visit count $T_{s,a}(t)$, define:*

$$Q_\Omega^{\text{ER}}(s, a) := Q_\Omega(s, a) + \frac{c_2}{T_{s,a}(t)}, \quad (26)$$

where $c_2 > 0$ is the ER strength parameter.

The ER-augmented E3W policy (which we call **E3W-ER**) is:

$$\pi_t^{\text{ER}}(a|s) = (1 - \lambda_s) \nabla \Omega^* \left(\frac{Q_\Omega^{\text{ER}}(s, \cdot)}{\tau} \right)(a) + \frac{\lambda_s}{|\mathcal{A}|}. \quad (27)$$

The backup operator remains unchanged: Q_Ω is updated using the original regularized backup, and the ER bonus is added only at *selection time*. This separation is important — the ER bonus affects *which actions are explored* but not *how values are estimated*.

6.1 ER-Augmented Policies for Each Regularizer

MENTS-ER. The softmax policy with ER bonus:

$$\pi^{\text{ER}}(a|s) \propto \exp\left(\frac{Q(s, a) + c_2/T_{s,a}}{\tau}\right). \quad (28)$$

The ER term acts as a temperature-scaled optimistic shift in the logits.

RENTS-ER. The prior-weighted policy with ER bonus:

$$\pi_t^{\text{ER}}(a|s) \propto \pi_{t-1}(a|s) \cdot \exp\left(\frac{Q_t(s, a) + c_2/T_{s,a}}{\tau}\right). \quad (29)$$

TENTS-ER (Tsallis + Effective Resistance). The sparse policy with ER bonus:

$$\pi^{\text{ER}}(a|s) = \max\left\{\frac{Q(s, a) + c_2/T_{s,a}}{\tau} - \frac{\sum_{a' \in \mathcal{K}_{\text{ER}}}(Q(s, a') + c_2/T_{s,a'})/\tau - 1}{|\mathcal{K}_{\text{ER}}|}, 0\right\}, \quad (30)$$

where $\mathcal{K}_{\text{ER}}$ is the sparsity set computed using the ER-augmented Q-values Q^{ER} .

Remark 23 (Adaptive sparsity in TENTS-ER). *The sparsity set $\mathcal{K}_{\text{ER}}$ is determined by which actions satisfy:*

$$1 + i \cdot \frac{Q(s, a_i) + c_2/T_{s,a_i}}{\tau} > \sum_{j=1}^i \frac{Q(s, a_j) + c_2/T_{s,a_j}}{\tau},$$

where a_i is the action with the i -th largest augmented Q-value. In the early phase, when many actions have small $T_{s,a}$, the ER bonus $c_2/T_{s,a}$ inflates Q-values broadly, causing more actions to satisfy the threshold and expanding $\mathcal{K}_{\text{ER}}$. As visits accumulate, $c_2/T_{s,a} \rightarrow 0$ and $\mathcal{K}_{\text{ER}} \rightarrow \mathcal{K}$ (the unaugmented sparsity set), contracting the policy back to sparse exploitation.

This provides an adaptive sparsity schedule: broad exploration early, sparse exploitation late — without any manual annealing. In contrast, vanilla TENTS has a fixed sparsity pattern determined solely by Q-values, which can prematurely exclude promising actions before they are sufficiently explored.

7 Convergence Analysis for Convex Regularized MCTS-ER

We derive the convergence properties of E3W-ER by following the proof architecture of Dam et al. [2021]. The key insight is that the ER perturbation $c_2/T_{s,a}$ vanishes at rate $O(\log t/t)$, which is absorbed by the exponential convergence envelope $\exp(-t/(\log t)^3)$. We establish this through a sequence of lemmas that parallel Lemmas 1–5 and Theorems 1–4 of Dam et al. [2021], explicitly tracking the ER perturbation at each step.

7.1 Leaf Concentration under E3W-ER

Lemma 24 (ER-modified Lemma 1 — Leaf concentration). *For the E3W-ER policy applied to a tree, let $\hat{r}$ and r be respectively the average and the expected reward at the leaf node, with the reward distribution being σ^2 -sub-Gaussian. For $t \geq 4$:*

$$\mathbb{P}\left(\|r - \hat{r}_t\|_\infty \geq \frac{2\sigma}{\log(2+t)}\right) \leq 4|\mathcal{A}| \exp\left(-\frac{t}{(\log(2+t))^3}\right).$$

Proof. At the leaf nodes, the ER bonus does not affect the playout evaluation — rewards are still collected as i.i.d. samples from the playout policy π_0 . The E3W-ER policy at the leaf differs from E3W only in the selection phase, which determines *how often* each action is visited, not *how rewards are generated*.

Let $N_t(a)$ be the number of times action a has been chosen until time t , and $\hat{N}_t(a) = \sum_{s=1}^t \pi_s^{\text{ER}}(a)$. The E3W-ER policy (27) contains the uniform mixing term $\lambda_s/|\mathcal{A}|$ with $\lambda_s = |\mathcal{A}|\epsilon/\log(1+s)$. Therefore, for all a and $t \geq 4$:

$$\hat{N}_t(a) = \sum_{s=1}^t \pi_s^{\text{ER}}(a) \geq \sum_{s=1}^t \frac{\lambda_s}{|\mathcal{A}|} = \sum_{s=1}^t \frac{\epsilon}{\log(1+s)} \geq \frac{t}{2 \log(2+t)},$$

where the last inequality follows from the integral bound $\sum_{s=1}^t 1/\log(1+s) \geq \int_1^{1+t} 1/\log(1+s) ds \geq t/\log(2+t) - 1/\log 2$.

From Theorem 2.19 of Dam et al. [2021, Wainwright, 2019], the concentration inequality for martingale differences gives:

$$\mathbb{P}(|N_t(a) - \hat{N}_t(a)| > \delta) \leq 2 \exp\left(-\frac{2\delta^2}{t}\right),$$

where $\sigma_s^2 \leq 1/4$ is the variance of the Bernoulli with $p = \pi_s^{\text{ER}}(a)$. Define the event $\mathcal{E} = \{\forall a \in \mathcal{A} : |N_t(a) - \hat{N}_t(a)| \leq \delta\}$. By a union bound:

$$\mathbb{P}(|\hat{N}_t(a) - N_t(a)| \geq \delta) \leq 2|\mathcal{A}| \exp\left(-\frac{2\delta^2}{t}\right). \quad (31)$$

Conditioned on $\mathcal{E}$ with $\delta = t/(4\log(2+t))$, we get $N_t(a) \geq t/(4\log(2+t))$. For any action a , by the sub-Gaussian property:

$$\mathbb{P}\left(|r(a) - \hat{r}_t(a)| > \sqrt{\frac{8\sigma^2 \log(2/\delta') \log(2+t)}{t}}\right) \leq \mathbb{P}\left(|r(a) - \hat{r}_t(a)| > \sqrt{\frac{2\sigma^2 \log(2/\delta')}{N_t(a)}}\right) \leq \delta'.$$

Choosing δ' such that $\log(2/\delta') = 1/(\log(2+t))^3$:

$$\mathbb{P}\left(|r(a) - \hat{r}_t(a)| > \frac{2\sigma}{\log(2+t)}\right) \leq 2 \exp\left(-\frac{1}{(\log(2+t))^3}\right).$$

Combining with (31) via the law of total probability:

$$\begin{aligned} \mathbb{P}\left(\|r - \hat{r}_t\|_\infty > \frac{2\sigma}{\log(2+t)}\right) &\leq \mathbb{P}\left(\|r - \hat{r}_t\|_\infty > \frac{2\sigma}{\log(2+t)} \mid \mathcal{E}\right) + \mathbb{P}(\mathcal{E}^c) \\ &\leq 2|\mathcal{A}| \exp\left(-\frac{1}{(\log(2+t))^3}\right) + 2|\mathcal{A}| \exp\left(-\frac{t}{(\log(2+t))^3}\right) \\ &\leq 4|\mathcal{A}| \exp\left(-\frac{t}{(\log(2+t))^3}\right). \quad \square \end{aligned}$$

7.2 Visit Count Concentration under E3W-ER

Lemma 25 (ER-modified Lemma 2 — Lipschitz policy perturbation). *Given two Q -value vectors $Q^{(1)}$ and $Q^{(2)}$, the E3W-ER policies satisfy: there exists $L > 0$ such that*

$$\|\nabla\Omega^*(Q_{\text{ER}}^{(1)}/\tau) - \nabla\Omega^*(Q_{\text{ER}}^{(2)}/\tau)\|_p \leq L\|Q_{\text{ER}}^{(1)}/\tau - Q_{\text{ER}}^{(2)}/\tau\|_p.$$

Proof. This follows directly from the Lipschitz continuity of $\nabla\Omega^*$ (Proposition 19). Since the ER bonus is added before applying $\nabla\Omega^*$, the Lipschitz property applies to the augmented Q -values with the same constant L . The ℓ_p -norm depends on the choice of regularizer: $p = \infty$ for maximum and relative entropy, and $p = 2$ for Tsallis entropy [cf. Dam et al., 2021]. $\square$

Lemma 26 (ER-modified Lemma 3 — Expected visit count concentration). *Consider E3W-ER applied to a tree. At any node s at depth d , define $N_t^*(s, a) = \pi^*(a|s) \cdot t$ and $\hat{N}_t(s, a) = \sum_{k=1}^t \pi_k^{\text{ER}}(a|s)$. There exist constants $C_{\text{ER}}^{(3)}, \hat{C}_{\text{ER}}^{(3)}$ such that*

$$\mathbb{P}\left(|\hat{N}_t(s, a) - N_t^*(s, a)| > \frac{C_{\text{ER}}^{(3)}t}{\log t}\right) \leq \hat{C}_{\text{ER}}^{(3)}|\mathcal{A}|t \exp\left(-\frac{t}{(\log t)^3}\right).$$

Proof. Define the event $\mathcal{E}_{r,k} = \{\|r(s', \cdot) - \hat{r}_k(s', \cdot)\|_\infty < 2\sigma / \log(2+k)\}$. Conditioned on $\bigcap_{i=1}^t \mathcal{E}_{r,i}$ and for $t \geq 4$, we bound $|\hat{N}_t(s, a) - N_t^*(s, a)|$:

$$\begin{aligned}
|\hat{N}_t(s, a) - N_t^*(s, a)| &\leq \sum_{k=1}^t |\pi_k^{\text{ER}}(a|s) - \pi^*(a|s)| + \sum_{k=1}^t \lambda_k \\
&\leq \sum_{k=1}^t \|\hat{\pi}_k^{\text{ER}}(\cdot|s) - \pi^*(\cdot|s)\|_\infty + \sum_{k=1}^t \lambda_k \\
&\leq \sum_{k=1}^t \|\hat{\pi}_k^{\text{ER}}(\cdot|s) - \pi^*(\cdot|s)\|_p + \sum_{k=1}^t \lambda_k \\
&\leq L \sum_{k=1}^t \|\hat{Q}_k^{\text{ER}}(s', \cdot)/\tau - Q^*(s', \cdot)/\tau\|_p + \sum_{k=1}^t \lambda_k, \tag{32}
\end{aligned}$$

where the last step uses Lemma 25. Now we decompose the Q-value perturbation:

$$\begin{aligned}
\|\hat{Q}_k^{\text{ER}}(s', \cdot)/\tau - Q^*(s', \cdot)/\tau\|_p &\leq \|\hat{Q}_k(s', \cdot)/\tau - Q^*(s', \cdot)/\tau\|_p + \|c_2/(\tau T_{s', \cdot}(k))\|_p \\
&\leq |\mathcal{A}|^{1/p} \gamma^d \|\hat{r}_k(s'', \cdot) - r(s'', \cdot)\|_\infty + \frac{c_2 |\mathcal{A}|^{1/p}}{\tau} \cdot \frac{4 \log(2+k)}{k},
\end{aligned}$$

where the first term uses the contraction property (Proposition 19) and the second uses the visit count lower bound $T_{s', a}(k) \geq k/(4 \log(2+k))$ from the E3W uniform mixing. Substituting into (32):

$$\begin{aligned}
|\hat{N}_t(s, a) - N_t^*(s, a)| &\leq L |\mathcal{A}|^{1/p} \gamma^d \sum_{k=1}^t \frac{2\sigma}{\log(2+k)} + \frac{4Lc_2 |\mathcal{A}|^{1/p}}{\tau} \sum_{k=1}^t \frac{\log(2+k)}{k} + \sum_{k=1}^t \frac{|\mathcal{A}| \epsilon}{\log(1+k)} \\
&\leq L |\mathcal{A}|^{1/p} \gamma^d \cdot \frac{2\sigma t}{\log t} + \frac{4Lc_2 |\mathcal{A}|^{1/p}}{\tau} \cdot (\log t)^2 + \frac{|\mathcal{A}| \epsilon t}{\log t} \\
&\leq \frac{C_{\text{ER}}^{(3)} t}{\log t},
\end{aligned}$$

for some constant $C_{\text{ER}}^{(3)}$ depending on $|\mathcal{A}|, p, d, \sigma, L, \gamma, c_2, \tau, \epsilon$. The crucial observation is that the ER contribution $\sum_{k=1}^t (\log k)^2/k = O((\log t)^2) = o(t/\log t)$, so it is absorbed into the leading $O(t/\log t)$ term. Finally:

$$\mathbb{P}\left(|\hat{N}_t(s, a) - N_t^*(s, a)| \geq \frac{C_{\text{ER}}^{(3)} t}{\log t}\right) \leq \sum_{i=1}^t \mathbb{P}(\mathcal{E}_{r,i}^c) \leq 4|\mathcal{A}|t \exp\left(-\frac{t}{(\log(2+t))^3}\right). \quad \square$$

Lemma 27 (ER-modified Lemma 4 — Actual visit count concentration). *Under E3W-ER, at any node s , let $N_t(s, a)$ be the actual number of times action a has been chosen until step t . There exist constants $C_{\text{ER}}^{(4)}, \hat{C}_{\text{ER}}^{(4)}$ such that*

$$\mathbb{P}\left(|N_t(s, a) - N_t^*(s, a)| > \frac{C_{\text{ER}}^{(4)} t}{\log t}\right) \leq \hat{C}_{\text{ER}}^{(4)} t \exp\left(-\frac{t}{(\log t)^3}\right).$$

Proof. Using Lemma 26 and the concentration bound (31):

$$\begin{aligned}
&\mathbb{P}\left(|N_t(s, a) - N_t^*(s, a)| > (1 + C_{\text{ER}}^{(3)}) \frac{t}{\log t}\right) \\
&\leq \mathbb{P}\left(|\hat{N}_t(s, a) - N_t^*(s, a)| > \frac{C_{\text{ER}}^{(3)} t}{\log t}\right) + \mathbb{P}\left(|N_t(s, a) - \hat{N}_t(s, a)| > \frac{t}{\log t}\right) \\
&\leq \hat{C}_{\text{ER}}^{(3)} |\mathcal{A}| t \exp\left(-\frac{t}{(\log(2+t))^3}\right) + 2|\mathcal{A}| \exp\left(-\frac{t}{(\log(2+t))^2}\right) \\
&\leq \hat{C}_{\text{ER}}^{(4)} t \exp\left(-\frac{t}{(\log t)^3}\right). \quad \square
\end{aligned}$$

7.3 Exponential Convergence of ER-Augmented Value

Lemma 28 (ER perturbation bound). *Under E3W-ER, the ER bonus at any node s and action a satisfies: for $t \geq 4$,*

$$\frac{c_2}{T_{s,a}(t)} \leq \frac{4c_2 \log(2+t)}{t} =: \frac{C'_{\text{ER}} \log t}{t},$$

with probability at least $1 - \hat{C}_{\text{ER}}^{(4)} t \exp(-t/(\log t)^3)$.

Proof. By Lemma 27, when $N(s) = t$ is large enough, $N_t(s, a) \geq \frac{1}{2} \pi^*(a|s)t$ with the stated probability. Using the uniform mixing lower bound from the E3W component: $T_{s,a}(t) \geq t/(4 \log(2+t))$, which gives $c_2/T_{s,a}(t) \leq 4c_2 \log(2+t)/t$. $\square$

Lemma 29 (Lipschitz stability of regularized values). *For any strongly convex regularizer Ω :*

$$|V_{\Omega}^{\text{ER}}(s) - V_{\Omega}(s)| = |\Omega^*(Q_{\Omega}^{\text{ER}}/\tau) - \Omega^*(Q_{\Omega}/\tau)| \leq \frac{c_2}{\tau \cdot \min_a T_{s,a}(t)}. \quad (33)$$

Proof. By the contraction property (Proposition 19):

$$|\Omega^*(Q^{\text{ER}}/\tau) - \Omega^*(Q/\tau)| \leq \|Q^{\text{ER}}/\tau - Q/\tau\|_{\infty} = \frac{1}{\tau} \max_a \frac{c_2}{T_{s,a}(t)} = \frac{c_2}{\tau \cdot \min_a T_{s,a}(t)}. \quad \square$$

We now define the visit count event analogous to Lemma 5 of Dam et al. [2021]:

Definition 30 (Visit count event). *At node s , define $N^*(s, a) = \pi^*(a|s)N(s)$ and the event*

$$\mathcal{E}_s := \left\{ \forall a \in \mathcal{A} : |N(s, a) - N^*(s, a)| < \frac{N^*(s, a)}{2} \right\}.$$

Lemma 31 (ER-modified Lemma 5 — Conditional value concentration). *At any node s of the tree, for $\epsilon > 0$:*

$$\mathbb{P}(|V_{\Omega}(s) - V_{\Omega}^*(s)| > \epsilon \mid \mathcal{E}_s) \leq C_{\text{ER}}^{(5)} \exp \left\{ -\frac{N(s)}{\hat{C}_{\text{ER}}^{(5)} \sigma(\log(2 + N(s)))^2} \right\}.$$

Proof. We prove by induction on the depth D .

Base case ($D = 1$). From the contraction property:

$$|V_{\Omega}(s) - V_{\Omega}^*(s)| = \|\Omega^*(Q_{\Omega}(s, \cdot)) - \Omega^*(Q_{\Omega}^*(s, \cdot))\|_{\infty} \leq \gamma \|\hat{r} - r^*\|_{\infty}.$$

By Lemma 24 with $\epsilon = 2\sigma\gamma/\log(2 + N(s))$:

$$\begin{aligned} \mathbb{P}(|V_{\Omega}(s) - V_{\Omega}^*(s)| > \epsilon) &\leq \mathbb{P}(\gamma \|\hat{r} - r^*\|_{\infty} > \epsilon) \\ &\leq 4|\mathcal{A}| \exp \left\{ -\frac{N(s)}{2\sigma\gamma(\log(2 + N(s)))^2} \right\} \\ &= C_{\text{ER}}^{(5)} \exp \left\{ -\frac{N(s)}{\hat{C}_{\text{ER}}^{(5)} (\log(2 + N(s)))^2} \right\}. \end{aligned}$$

Inductive step ($D - 1 \rightarrow D$). Assume the bound holds at depth $D - 1$. Define $V_{\Omega}(s_a) = Q_{\Omega}(s, a)$ where s_a is the state reached by taking action a from s . At depth $D - 1$:

$$\mathbb{P}(|V_{\Omega}(s_a) - V_{\Omega}^*(s_a)| > \epsilon) \leq C_a \exp \left\{ -\frac{N(s_a)}{\hat{C}_a (\log(2 + N(s_a)))^2} \right\}.$$

At depth D , by the contraction property:

$$|V_{\Omega}(s) - V_{\Omega}^*(s)| \leq \gamma \|Q_{\Omega}(s, \cdot) - Q_{\Omega}^*(s, \cdot)\|_{\infty} = \gamma |Q_{\Omega}(s, a) - Q_{\Omega}^*(s, a)|,$$

for some action a achieving the ∞ -norm. Therefore:

$$\begin{aligned}\mathbb{P}(|V_\Omega(s) - V_\Omega^*(s)| > \epsilon) &\leq \mathbb{P}(\gamma \|Q_\Omega(s, a) - Q_\Omega^*(s, a)\| > \epsilon) \\ &\leq C_a \exp\left\{-\frac{N(s_a)}{\hat{C}_a(\log(2 + N(s_a)))^2}\right\} \\ &\leq C_a \exp\left\{-\frac{N(s_a)}{\hat{C}_a(\log(2 + N(s)))^2}\right\}.\end{aligned}$$

From the induction hypothesis, $\lim_{t \rightarrow \infty} N(s_a) = \infty$. By Lemma 27, when $N(s)$ is large enough, conditioned on $\mathcal{E}_s$: $N(s_a) \geq \frac{1}{2}\pi^*(a|s)N(s)$. Therefore we can find $C_{\text{ER}}^{(5)}, \hat{C}_{\text{ER}}^{(5)}$ such that

$$\mathbb{P}(|V_\Omega(s) - V_\Omega^*(s)| > \epsilon \mid \mathcal{E}_s) \leq C_{\text{ER}}^{(5)} \exp\left\{-\frac{N(s)}{\hat{C}_{\text{ER}}^{(5)}(\log(2 + N(s)))^2}\right\}. \quad \square$$

Theorem 32 (Exponential convergence of ER-augmented regularized MCTS). *At the root node s with $N(s)$ visitations, for any strongly convex regularizer Ω and $\epsilon > 0$:*

$$\mathbb{P}(|V_\Omega^{\text{ER}}(s) - V_\Omega^*(s)| > \epsilon) \leq C_{\text{ER}} \exp\left\{-\frac{N(s)}{\hat{C}_{\text{ER}}\sigma \log^2(2 + N(s))}\right\}.$$

Proof. We decompose:

$$|V_\Omega^{\text{ER}}(s) - V_\Omega^*(s)| \leq \underbrace{|V_\Omega^{\text{ER}}(s) - V_\Omega(s)|}_{E_1} + \underbrace{|V_\Omega(s) - V_\Omega^*(s)|}_{E_2}.$$

Bounding E_1 . By Lemmas 29 and 28:

$$E_1 \leq \frac{c_2}{\tau \cdot \min_a T_{s,a}(t)} \leq \frac{4c_2 \log(2 + N(s))}{\tau \cdot N(s)} = \frac{C'_{\text{ER}} \log N(s)}{\tau N(s)}.$$

This is $O(\log t/t)$. For any fixed $\epsilon > 0$, there exists $N_0(\epsilon)$ such that $E_1 < \epsilon/2$ for all $N(s) \geq N_0(\epsilon)$.

Bounding E_2 . We verify the three ingredients of the inductive proof:

(i) Leaf concentration. Lemma 24 holds unchanged since the ER bonus does not affect payout evaluation.

(ii) Visit count concentration. Lemma 27 establishes $|N_t(s, a) - N_t^*(s, a)| = O(t/\log t)$ under E3W-ER, with the ER contribution absorbed as $O((\log t)^2) = o(t/\log t)$.

(iii) Inductive contraction. At depth D : $|V_\Omega(s) - V_\Omega^*(s)| \leq \gamma \|Q_\Omega(s, \cdot) - Q_\Omega^*(s, \cdot)\|_\infty$. Since the backup uses Q_Ω (without ER — the ER bonus is only injected at selection time), the contraction holds exactly with constant γ .

By Lemma 31, E_2 satisfies the exponential bound with modified constants. Combining: for $N(s) \geq N_0(\epsilon)$,

$$\mathbb{P}(|V_\Omega^{\text{ER}}(s) - V_\Omega^*(s)| > \epsilon) \leq \mathbb{P}(E_2 > \epsilon/2) \leq C_{\text{ER}}^{(5)} \exp\left\{-\frac{N(s)}{\hat{C}_{\text{ER}}^{(5)}\sigma(\log(2 + N(s)))^2}\right\}.$$

For $N(s) < N_0(\epsilon)$, the bound holds trivially by choosing C_{ER} large enough. $\square$

7.4 Exponential Action Convergence

Theorem 33 (Exponential action convergence of E3W-ER). *Let a_t be the action returned by E3W-ER at step t . For t large enough:*

$$\mathbb{P}(a_t \neq a^*) \leq C_{\text{ER}} t \exp\left\{-\frac{t}{\hat{C}_{\text{ER}}\sigma(\log t)^3}\right\}.$$

Proof. Let a^* be the action with the largest value estimate at root s . The probability that E3W-ER selects a sub-optimal arm is:

$$\mathbb{P}(a_t \neq a^*) \leq \sum_a \mathbb{P}(V_\Omega^{\text{ER}}(s_a) > V_\Omega^{\text{ER}}(s_{a^*}) \mid \mathcal{E}_s) + \mathbb{P}(\mathcal{E}_s^c).$$

For the first term, define $\Delta^{\text{ER}} = V_\Omega^*(s_{a^*}) - V_\Omega^*(s_a)$. Then:

$$\begin{aligned} & \mathbb{P}(V_\Omega^{\text{ER}}(s_a) > V_\Omega^{\text{ER}}(s_{a^*}) \mid \mathcal{E}_s) \\ &= \mathbb{P}((V_\Omega^{\text{ER}}(s_a) - V_\Omega^*(s_a)) - (V_\Omega^{\text{ER}}(s_{a^*}) - V_\Omega^*(s_{a^*})) \geq \Delta^{\text{ER}} \mid \mathcal{E}_s). \end{aligned}$$

Now $V_\Omega^{\text{ER}}(s_a) = V_\Omega(s_a) + O(\log t/t)$ by Lemma 29. For t large enough that the ER perturbation is less than $\Delta^{\text{ER}}/4$ at both s_a and s_{a^*} , the effective gap is at least $\Delta^{\text{ER}}/2$. With $\alpha + \beta = 1$, $\alpha, \beta > 0$:

$$\begin{aligned} & \mathbb{P}(V_\Omega^{\text{ER}}(s_a) > V_\Omega^{\text{ER}}(s_{a^*}) \mid \mathcal{E}_s) \\ & \leq \mathbb{P}(|V_\Omega(s_a) - V_\Omega^*(s_a)| \geq \alpha \Delta^{\text{ER}}/2 \mid \mathcal{E}_s) + \mathbb{P}(|V_\Omega(s_{a^*}) - V_\Omega^*(s_{a^*})| \geq \beta \Delta^{\text{ER}}/2 \mid \mathcal{E}_s) \\ & \leq C_a \exp \left\{ -\frac{N(s_a)(\alpha \Delta^{\text{ER}}/2)}{\hat{C}_a(\log(2 + N(s)))^2} \right\} + C_{a^*} \exp \left\{ -\frac{N(s_{a^*})(\beta \Delta^{\text{ER}}/2)}{\hat{C}_{a^*}(\log(2 + N(s)))^2} \right\}, \end{aligned}$$

where we applied Lemma 31. Define $1/\hat{C} = \min\{(\alpha \Delta^{\text{ER}}/2)/\hat{C}_a, (\beta \Delta^{\text{ER}}/2)/\hat{C}_{a^*}\}$ and $C = \max\{C_a, C_{a^*}\}/|\mathcal{A}|$:

$$\sum_a \mathbb{P}(V_\Omega^{\text{ER}}(s_a) > V_\Omega^{\text{ER}}(s_{a^*}) \mid \mathcal{E}_s) \leq C \exp \left\{ -\frac{t}{\hat{C}\sigma(\log(2 + t))^2} \right\}.$$

For $\mathbb{P}(\mathcal{E}_s^c)$, by Lemma 27: $\mathbb{P}(\mathcal{E}_s^c) \leq \hat{C}_{\text{ER}}^{(4)} t \exp(-t/(\log t)^3)$. Therefore:

$$\mathbb{P}(a_t \neq a^*) \leq C \exp \left\{ -\frac{t}{\hat{C}\sigma(\log(2 + t))^2} \right\} + \hat{C}_{\text{ER}}^{(4)} t \exp \left(-\frac{t}{(\log t)^3} \right) = O \left(t \exp \left\{ -\frac{t}{(\log t)^3} \right\} \right). \quad \square$$

7.5 Regret Analysis with ER Bonus

Theorem 34 (Regret of E3W-ER). *Consider E3W-ER applied to the tree. Let $D_{\Omega^*}(x, y) = \Omega^*(x) - \Omega^*(y) - \nabla \Omega^*(y)(x - y)$ be the Bregman divergence. The expected pseudo-regret $R_n = nV^* - \sum_{i=1}^n V_{i_t}$ satisfies:*

$$\mathbb{E}[R_n] \leq -\tau \Omega(\hat{\pi}) + \sum_{t=1}^n D_{\Omega^*}(\hat{V}^t + V, \hat{V}^t) + O \left(\frac{n}{\log n} \right) + c_2 H_n, \quad (34)$$

where $H_n = \sum_{t=1}^n \sum_a \frac{\pi_t^{\text{ER}}(a|s)}{T_{s,a}(t)}$ is the cumulative ER exploration cost.

Proof. Without loss of generality, assume $V_i \in [-1, 0]$ for all $i \in [1, |\mathcal{A}|]$. By definition of regret:

$$\mathbb{E}[R_n] = nV^* - \sum_{t=1}^n \langle \hat{\pi}_t^{\text{ER}}(\cdot), V(\cdot) \rangle \leq \hat{V}^1(0) - \sum_{t=1}^n \langle \hat{\pi}_t^{\text{ER}}(\cdot), V(\cdot) \rangle \leq -\tau \Omega(\hat{\pi}) - \sum_{t=1}^n \langle \hat{\pi}_t^{\text{ER}}(\cdot), V(\cdot) \rangle.$$

Under E3W-ER, the policy acts on augmented Q-values. Decompose:

$$\langle \hat{\pi}_t^{\text{ER}}(\cdot), V(\cdot) \rangle = \langle \hat{\pi}_t^{\text{ER}}(\cdot), Q^{\text{ER}}(\cdot) \rangle - c_2 \sum_a \frac{\hat{\pi}_t^{\text{ER}}(a|s)}{T_{s,a}(t)} + \langle \hat{\pi}_t^{\text{ER}}(\cdot), V(\cdot) - Q(\cdot) \rangle.$$

For the first and third terms, by the definition of the E3W-ER tree policy:

$$\begin{aligned} -\sum_{t=1}^n \langle \hat{\pi}_t^{\text{ER}}(\cdot), V(\cdot) \rangle &= -\sum_{t=1}^n \langle (1 - \lambda_t) \nabla \Omega^*(\hat{V}^t(\cdot)), V(\cdot) \rangle - \sum_{t=1}^n \left\langle \frac{\lambda_t(\cdot)}{|\mathcal{A}|}, V(\cdot) \right\rangle + c_2 H_n \\ &\leq -\sum_{t=1}^n \langle \nabla \Omega^*(\hat{V}^t(\cdot)), V(\cdot) \rangle + O\left(\frac{n}{\log n}\right) + c_2 H_n, \end{aligned}$$

where $\sum_{k=1}^n \lambda_k / |\mathcal{A}| = \sum_{k=1}^n \epsilon / \log(1 + k) = O(n / \log n)$. Now, using the Bregman divergence:

$$\begin{aligned} -\sum_{t=1}^n \langle \nabla \Omega^*(\hat{V}^t), V \rangle &= \sum_{t=1}^n [\Omega^*(\hat{V}^t + V) - \Omega^*(\hat{V}^t) - \langle \nabla \Omega^*(\hat{V}^t), V \rangle] \\ &\quad - \sum_{t=1}^n [\Omega^*(\hat{V}^t + V) - \Omega^*(\hat{V}^t)] \\ &= \sum_{t=1}^n D_{\Omega^*}(\hat{V}^t + V, \hat{V}^t) - \sum_{t=1}^n [\Omega^*(\hat{V}^t + V) - \Omega^*(\hat{V}^t)] \\ &\leq \sum_{t=1}^n D_{\Omega^*}(\hat{V}^t + V, \hat{V}^t) + n \|V\|_{\infty} \\ &\leq \sum_{t=1}^n D_{\Omega^*}(\hat{V}^t + V, \hat{V}^t), \end{aligned}$$

where the penultimate step uses the contraction property, and the last step uses $V_i \leq 0$. The cumulative ER cost is bounded: since $\sum_{t=1}^n 1/T_{s,a}(t) \leq 1 + \log n$ for each action (by the harmonic series bound), $H_n \leq |\mathcal{A}|(1 + \log n) = O(|\mathcal{A}| \log n)$. $\square$

Specializing to each regularizer, using the Bregman divergence bounds from Dam et al. [2021]:

Corollary 35 (MENTS-ER regret). *For maximum entropy ($\Omega(\pi) = \sum_a \pi \log \pi$): $D_{\Omega^*}(\hat{V}^t + V, \hat{V}^t) \leq |\mathcal{A}|/\tau$. Therefore:*

$$\mathbb{E}[R_n] \leq \tau \log |\mathcal{A}| + n|\mathcal{A}|/\tau + O(n/\log n) + O(c_2 |\mathcal{A}| \log n).$$

Corollary 36 (RENTS-ER regret). *For relative entropy ($\Omega(\pi_t) = D_{\text{KL}}(\pi_t \| \pi_{t-1})$), with $m = \min_a \pi_{t-1}(a|s)$:*

$$\mathbb{E}[R_n] \leq \tau(\log |\mathcal{A}| - 1/m) + n|\mathcal{A}|/\tau + O(n/\log n) + O(c_2 |\mathcal{A}| \log n).$$

Corollary 37 (TENTS-ER regret). *For Tsallis entropy ($\Omega(\pi) = \frac{1}{2}(\|\pi\|_2^2 - 1)$), by Dam et al. [2021]: $D_{\Omega^*}(\hat{V}^t + V, \hat{V}^t) \leq |\mathcal{K}_{\text{ER}}|/2$. Therefore:*

$$\mathbb{E}[R_n] \leq \tau(|\mathcal{A}| - 1)/|\mathcal{A}| + n|\mathcal{K}_{\text{ER}}|/2 + O(n/\log n) + O(c_2 |\mathcal{A}| \log n),$$

where $|\mathcal{K}_{\text{ER}}|$ is the effective sparsity set size under ER-augmented Q -values.

Remark 38 (TENTS-ER regret advantage). *The TENTS-ER bound features $|\mathcal{K}_{\text{ER}}|/2$ rather than $|\mathcal{A}|/\tau$, which becomes tighter for increasing $|\mathcal{A}|$. Since $|\mathcal{K}_{\text{ER}}| \rightarrow |\mathcal{K}|$ as $t \rightarrow \infty$ and $|\mathcal{K}| \ll |\mathcal{A}|$ in practice, TENTS-ER preserves the advantage of Tsallis entropy in high-branching-factor problems. The ER cost $O(c_2 |\mathcal{A}| \log n)$ is dominated by $O(n/\log n)$ and does not affect the leading-order regret.*

7.6 Error Analysis with ER Bonus

Theorem 39 (Error of ER-augmented regularized value). *For any $\delta > 0$ and strongly convex Ω , define $\Psi_{\text{ER}} = \sqrt{\hat{C}_{\text{ER}} \sigma^2 \log(C_{\text{ER}}/\delta)/(2N(s))}$. The error $\varepsilon_{\Omega}^{\text{ER}} = V_{\Omega}^{\text{ER}}(s) - V^*(s)$ satisfies, with probability at least $1 - \delta$:*

$$-\Psi_{\text{ER}} - \frac{\tau(U_{\Omega} - L_{\Omega}^{\min})}{1 - \gamma} \leq \varepsilon_{\Omega}^{\text{ER}} \leq \Psi_{\text{ER}} + \frac{c_2}{\tau \cdot \min_a T_{s,a}}.$$

Proof. From Theorem 32, define $\delta = C_{\text{ER}} \exp\{-2N(s)\epsilon^2/(\hat{C}_{\text{ER}}\sigma^2)\}$, so $\epsilon = \Psi_{\text{ER}}$. For any $\delta > 0$, with probability at least $1 - \delta$:

$$|V_{\Omega}(s) - V_{\Omega}^*(s)| \leq \Psi_{\text{ER}}.$$

This gives: $-\Psi_{\text{ER}} \leq V_{\Omega}(s) - V_{\Omega}^*(s) \leq \Psi_{\text{ER}}$, hence

$$-\Psi_{\text{ER}} + V_{\Omega}^*(s) \leq V_{\Omega}(s) \leq \Psi_{\text{ER}} + V_{\Omega}^*(s).$$

By the boundedness of regularized MDPs [Geist et al., 2019, Theorem 2]: $V^*(s) - \frac{\tau(U_{\Omega} - L_{\Omega}^{\min})}{1 - \gamma} \leq V_{\Omega}^*(s) \leq V^*(s)$. Therefore:

$$-\Psi_{\text{ER}} + V^*(s) - \frac{\tau(U_{\Omega} - L_{\Omega}^{\min})}{1 - \gamma} \leq V_{\Omega}(s) \leq \Psi_{\text{ER}} + V^*(s).$$

Adding the ER perturbation from Lemma 29: $V_{\Omega}^{\text{ER}}(s) = V_{\Omega}(s) + c_2/(\tau \min_a T_{s,a})$, so:

$$-\Psi_{\text{ER}} - \frac{\tau(U_{\Omega} - L_{\Omega}^{\min})}{1 - \gamma} \leq V_{\Omega}^{\text{ER}}(s) - V^*(s) \leq \Psi_{\text{ER}} + \frac{c_2}{\tau \cdot \min_a T_{s,a}},$$

where $c_2/(\tau \min_a T_{s,a}) = O(\log N(s)/N(s)) \rightarrow 0$ asymptotically by Lemma 28. $\square$

Specializing using the entropy bounds from Dam et al. [2021], Lee et al. [2018]: for maximum entropy, $-\log |\mathcal{A}| \leq \Omega(\pi) \leq 0$; for relative entropy with $m = \min_a \pi_{t-1}(a|s)$, $0 \leq \Omega(\pi) \leq -\log |\mathcal{A}| + \log(1/m)$; for Tsallis entropy, $-(|\mathcal{A}| - 1)/(2|\mathcal{A}|) \leq \Omega(\pi) \leq 0$.

Corollary 40 (MENTS-ER error). $-\Psi_{\text{ER}} - \tau \log |\mathcal{A}|/(1 - \gamma) \leq \varepsilon_{\Omega}^{\text{ER}} \leq \Psi_{\text{ER}} + O(\log t/t)$.

Corollary 41 (RENTS-ER error). $-\Psi_{\text{ER}} - \tau(\log |\mathcal{A}| - \log(1/m))/(1 - \gamma) \leq \varepsilon_{\Omega}^{\text{ER}} \leq \Psi_{\text{ER}} + O(\log t/t)$.

Corollary 42 (TENTS-ER error). $-\Psi_{\text{ER}} - (|\mathcal{A}| - 1)\tau/(2|\mathcal{A}|(1 - \gamma)) \leq \varepsilon_{\Omega}^{\text{ER}} \leq \Psi_{\text{ER}} + O(\log t/t)$.

Remark 43 (TENTS-ER retains tightest error bounds). *As in the non-ER case, TENTS-ER enjoys the smallest error among the three regularizers when $|\mathcal{A}|$ is large, because $(|\mathcal{A}| - 1)/(2|\mathcal{A}|) < \log |\mathcal{A}|$ for all $|\mathcal{A}| \geq 2$. The ER perturbation adds only $O(\log t/t)$ to the upper bound, negligible compared to $O(1/\sqrt{N(s)})$.*

7.7 Summary of All MCTS-ER Variants

Table 2: Convergence properties for all MCTS-ER variants. All ER-augmented variants preserve the rates of their base algorithms.

Algorithm	Conv. Rate	Regret (leading)	Error (lower)	Sparsity
Power-UCT-ER	$O(n^{-1/2})$	—	$O(n^{-1/2})$	No
MENTS-ER	$\exp(-t/(\log t)^3)$	$n \mathcal{A} /\tau$	$\tau \log \mathcal{A} /(1 - \gamma)$	No
RENTS-ER	$\exp(-t/(\log t)^3)$	$n \mathcal{A} /\tau$	$\tau(\log \mathcal{A} - \log m)/(1 - \gamma)$	No
TENTS-ER	$\exp(-t/(\log t)^3)$	$n \mathcal{K}_{\text{ER}} /2$	$\tau(\mathcal{A} - 1)/(2 \mathcal{A} (1 - \gamma))$	Yes

8 Discussion and Extensions

8.1 When Does the ER Bonus Help?

The ER bonus $C_2/T_{s,a}$ provides the most benefit in the *sparse planning regime* where the number of simulations n is small relative to the tree size. In this regime, many edges have small visit counts, and the ER bonus provides:

1. **Path-level uncertainty awareness.** A node deep in a well-explored branch may have lower total bonus than a shallow node in a rarely-visited branch. The ER contribution captures this because $R_{\text{eff}}(\text{root}, s) = \sum_{e \in \text{path}} 1/N(e)$ accumulates uncertainty along the entire trajectory.
2. **Bottleneck detection.** A single weakly-visited edge in an otherwise well-explored trajectory creates a large ER contribution, redirecting exploration to reinforce that bottleneck. Standard UCB bonuses at non-bottleneck nodes would not detect this.
3. **Complementarity with power mean.** The power mean backup with $p > 1$ provides an optimistic value estimate. The ER bonus provides an optimistic exploration bonus based on structural uncertainty. Together, they address both value uncertainty and trajectory uncertainty.

8.2 Extension to State Graphs with Transpositions

When the underlying MDP has state transpositions (multiple action sequences reaching the same state), the search tree does not capture the full graph structure. If we maintain a state graph G alongside the search tree, recording all discovered transitions including transpositions, then $R_{\text{eff}}(\text{root}, s; G)$ becomes a non-trivial spectral quantity that accounts for parallel paths.

However, the convergence analysis breaks down in this setting because the ER bonus becomes a *global* quantity that depends on the entire graph structure. Specifically:

- The local decomposition (Remark 3) no longer holds.
- Adding a transposition edge changes R_{eff} for all nodes simultaneously.
- The node-by-node bandit decomposition used in the inductive proof (Theorem 15) is invalidated.

A potential approach for the transposition case is to use a *potential function* argument based on the total effective resistance $\Phi_n = \sum_s R_{\text{eff}}(\text{root}, s; G_n)$, which is monotonically decreasing by Rayleigh monotonicity. We leave this as an open problem for future work.

8.3 Connection to Power Mean Resistance

The effective resistance has an intriguing connection to the power mean parameter p . For parallel resistors, the combined resistance is the harmonic mean ($p = -1$ power mean) of individual resistances. This suggests a generalized notion of *power mean resistance*:

$$R_{\text{eff}}^{(q)}(s, t) = \left(\sum_{\text{paths } \pi: s \rightarrow t} R(\pi)^q \right)^{1/q},$$

where $R(\pi)$ is the resistance of path π . The case $q = -1$ recovers the standard effective resistance (harmonic mean over paths), $q = -\infty$ gives the shortest path, and $q = +\infty$ gives the longest path.

Combining this with the power mean backup parameter p of MCTS-ER creates a two-parameter family that controls both *value aggregation* (via p) and *path aggregation* (via q), offering a rich framework for risk-sensitive planning.

9 Conclusion

We have introduced MCTS-ER, a family of MCTS algorithms augmented with an effective resistance bonus that captures trajectory-level uncertainty.

In Part I, we showed that augmenting Stochastic-Power-UCT with an additive ER bonus preserves the $O(n^{-1/2})$ polynomial convergence rate, with the ER bonus asymptotically dominated by the polynomial exploration term.

In Part II, we showed that augmenting convex-regularized MCTS (MENTS, RENTS, TENTS) with ER-augmented Q-values preserves the exponential convergence rate $O(t \exp(-t/(\log t)^3))$. We derived regret bounds and error analysis for each regularizer, showing that the ER cost appears only as lower-order $O(c_2 \log n)$ terms. The most compelling combination is TENTS-ER, where the ER bonus dynamically modulates the Tsallis sparsity set $\mathcal{K}$, providing an adaptive exploration schedule — broad early, sparse late — that neither Tsallis entropy nor ER achieves alone.

Across both frameworks, the ER bonus comes “for free” in terms of asymptotic rates while providing structurally motivated finite-time improvements through path-level uncertainty awareness. The extension to state graphs with transpositions, where effective resistance becomes a non-trivial spectral quantity with parallel-path effects, remains an important open direction.

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